

Project Title: *Automated Assessment of Carbon Sequestration in Armenian Forest Stands: A Macro-Based ImageJ Analysis of Publicly Available Satellite Data.*

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1. Abstract

Armenia's mountainous forests are vital carbon sinks, yet they face increasing pressure from regional warming and moisture stress. This project proposes a non-invasive, automated workflow using **ImageJ/Fiji** to quantify **Leaf Area Index (LAI)** and biomass from publicly available **Sentinel-2 multispectral imagery**. By focusing on two distinct ecological zones, the humid northern forests of Dilijan and the drought-vulnerable southern stands of Syunik, the study demonstrates how open-source digital tools can replace expensive fieldwork. The outcome is a validated ImageJ Macro that automates the conversion of spectral data into CO₂ sequestration estimates, supporting Armenia's national climate adaptation goals.

2. Objectives

- **Automation:** Develop an ImageJ Macro to batch-process multispectral satellite images.
- **Quantification:** Calculate the **Normalized Difference Vegetation Index (NDVI)** as a proxy for canopy health.
- **Carbon Modeling:** Estimate Above-Ground Biomass (AGB) and CO₂ equivalents (CO_{2e}) for Armenian broadleaf species (*Fagus orientalis* and *Quercus macranthera*).
- **Comparative Analysis:** Evaluate the carbon-density variance between northern and southern Armenian forest clusters.

3. Methodology

3.1 Data Acquisition

Publicly available 10-meter resolution imagery will be sourced from the **European Space Agency (ESA) Sentinel-2** mission via the **Sentinel-Hub EO Browser**.

- **Target Bands:** Band 4 (Red, $\approx 655\text{ nm}$) and Band 8 (Near-Infrared, $\approx 842\text{ nm}$).
- **Criteria:** Less than 5% cloud cover, captured during the peak growing season (June–August).

3.2 ImageJ Macro Development

To ensure reproducibility and efficiency, the methodology utilizes a custom `.ijm` script. The macro automates the mathematical transformation of raw pixel values into environmental indicators.

Macro Logic & Pseudocode:

1. **Batch Import:** User selects a directory containing Red and NIR band pairs.
2. **32-bit Conversion:** Convert images to 32-bit to allow for decimal results in division.
3. **NDVI Calculation:**

$$NDVI = \frac{(Band\ 8 - Band\ 4)}{(Band\ 8 + Band\ 4)}$$

4. **Thresholding:** Apply a lower-bound threshold (>0.2) to filter out non-vegetated surfaces (soil/rocks).
5. **Calibration:** Apply a scaling factor to convert Mean NDVI to LAI.
6. **Results Export:** Append the Mean LAI, Area Fraction, and estimated Carbon value to a summary CSV.

3.3 Carbon Conversion Framework

The conversion follows a three-step mathematical bridge:

1. **LAI to AGB:** Based on Armenian forest studies, we apply the relation $AGB = \alpha \cdot LAI$, where α is a species-specific constant (e.g., $\approx 45\text{ t/ha}$ for mature Beech).
2. **Biomass to Carbon:** $Carbon \approx AGB \times 0.5$ (Carbon constitutes roughly half of dry biomass).
3. **Carbon to CO₂:** $CO_2\text{ sequestered} = Carbon \times 3.67$ (the ratio of the molecular weight of CO₂ to Carbon).

4. Spatiotemporal Analysis: Modeling the Seasonal Carbon Cycle

4.1 Comparative Temporal Framework

To quantify the "Carbon Breath" of Armenian forests, the methodology will analyze imagery across four distinct phenological stages:

1. **Peak Sequestration (June–August):** Maximum LAI and CO₂ absorption.
2. **Senescence (October–November):** Transition period for deciduous species (*Fagus* and *Quercus*).
3. **Dormancy (January–February):** Baseline carbon storage with minimal active sequestration.
4. **Rebirth (April–May):** Early leaf-out and rapid increase in photosynthetic activity.

4.2 Methodology for Seasonal Flux

- **Image Differencing:** Using ImageJ's [Process > Image Calculator \(Subtract\)](#), the dormant winter NDVI map will be subtracted from the peak summer NDVI map. This identifies the "**Net Annual Productivity**" of the forest stand.
- **Visualizing Phenology:** The "**Montage**" function in ImageJ ([Image > Stacks > Make Montage](#)) will be used to create a side-by-side visual timeline of a single forest plot (e.g., Dilijan) across the four seasons, providing a clear narrative of seasonal climate resilience.

4.3 Climate Resilience Indicators

By comparing the seasonal flux of the **Tavush** region (North) against the **Syunik** region (South), this project will identify which forests are "waking up" earlier or "shutting down" faster due to rising temperatures.

- **Early Leaf-out:** A marker of warming springs.
- **Early Senescence:** Often a marker of drought stress in southern Armenia, where trees drop leaves early to conserve water.

5. Expected Outcomes & Impact

- **Digital Map:** A visual "Heat Map" of carbon density across the Armenian highlands.
- **Policy Support:** Data-driven insights that can be used to argue for the protection of specific "High Carbon Value" areas in Syunik.
- **Scalability:** This workflow can be applied to any region in the Caucasus using only an internet connection and the developed ImageJ toolset.